

DecodeLabs Internship

Project 1: Data Cleaning & Preparation

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Objective

The objective of this project was to perform end-to-end data cleaning and validation using advanced Microsoft Excel keyboard shortcuts. The workflow prioritized rapid detection of missing values, zero-duplicate verification, date format normalization, and column-wide consistency checks to ensure 100% data readiness for downstream analytics.

Tools Used

- Microsoft Excel

Dataset Overview

The dataset contains customer order information with the following attributes:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

Data Cleaning Process

1. Missing Values

- Identified missing values in the CouponCode column.
- Replaced blank CouponCode values with "No coupon" to improve consistency.
- No missing values were found in the remaining columns.

2. Duplicate Records

- Checked the OrderID column for duplicate records.
- No duplicate Order IDs were found.

3. Data Format Validation

- Verified that all dates followed the **YYYY-MM-DD** format.
- Verified that CustomerID and OrderID followed a consistent format.

4. Numerical Data Validation

Validated the following numerical columns:

- Quantity
- UnitPrice
- ItemsInCart
- TotalPrice

Verified that TotalPrice correctly matched the calculation:

$$\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$$

5. Text Consistency

Verified consistent values in the following columns:

- Product
- PaymentMethod
- OrderStatus
- ReferralSource

No spelling inconsistencies were found.

Conclusion

All data cleaning and quality checks are complete. Missing entries were filled appropriately, no duplicate rows were found, and all dates and numbers are properly formatted. The dataset is accurate, consistent, and fully prepared for the next step.